SHASHI VARDHAN PENTHALA

Full Stack Developer | AIML Engineer

vardhans367@gmail.com +91 9063531983 pshashivardhan.vercel.app

linkedin.com/in/shashi-vardhan-a22358343 github.com/shashiv367 Hyderabad, India

Professional Summary

Full Stack Developer and AIML Engineer with hands-on experience building scalable web applications and AI-driven product features. Proficient in React.js, Spring Boot, Node.js, and Python, with applied expertise in natural language processing, machine learning model deployment, and microservices architecture. Skilled at designing secure RESTful APIs, optimizing database performance, and delivering measurable results including 25 percent faster page load times and 85 percent AI model accuracy. Seeking Full Stack Developer or AIML Engineer roles to apply engineering skills in building data-driven solutions.

Professional Experience

AIML Full Stack Developer Intern, *Grahmind Innovations Pvt Ltd* 12/2023 – 04/2024 | Hyderabad, India

- Developed and deployed AI-enabled web applications using React.js, Python, Flask, and MySQL, reducing average platform response time by 20 percent (from 400 milliseconds to 320 milliseconds) through query optimization
- Engineered a cross-platform Flutter mobile application spanning 10 screens with shared REST API backend, maintaining 99.9 percent crash-free user sessions across Android and iOS platforms.

Full Stack Developer Intern, *Nuhvin Global Services Private Limited* 05/2024 – 07/2026 | Hyderabad, India

- Spearheaded development of full-stack features using React.js and Spring Boot microservices, implementing code-splitting and lazy loading to reduce initial page load time by 25 percent (from 2.1 seconds to 1.6 seconds).
- Built and deployed MeetNow, a real-time video conferencing platform using Next.js, Node.js, and Mediasoup WebRTC SFU architecture, supporting 100 active daily users with 50 concurrent participants per session

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, HTML5, CSS3
AI and ML: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, TensorFlow, Pandas, NumPy
Frontend Technologies: React.js, Next.js, Redux, WebRTC, Responsive Web Design
Backend Technologies: Node.js, Express.js, Flask, FastAPI, RESTful APIs, JWT Authentication, OAuth
Databases: MySQL, PostgreSQL, MongoDB, Supabase, Database Optimization
DevOps and Cloud: Git, GitHub, AWS (EC2, S3, Lambda), CI/CD Pipelines, Jenkins
Other Technologies: Flutter, Mediasoup, WebSocket, Postman, Power BI, Tableau

Education

Bachelor of Technology in Computer Science and Engineering 06/2023 – 05/2027 | Hyderabad, India
Malla Reddy University
CGPA: 8.79/10.0

Intermediate, State Board of Intermediate Education, *Deeksha Junior College* 2023 | Hyderabad, India
92.2 percent

High School, CBSE Board, *Unicent School* 2021 | Hyderabad, India
86.5 percent

Projects

Mental-Health AI Assistant (MINO), *Python, NLP, Flask, React.js, MongoDB, WebSocket*

- Developed an AI-driven real-time mental health chatbot using custom natural language processing pipelines including tokenization, named entity recognition, and sentiment analysis, achieving 85 percent accuracy in query classification
- Trained supervised learning models on a dataset of 15,000 labeled mental health conversations using Scikit-learn, achieving an F1 score of 0.84 through hyperparameter tuning and cross-validation
- Implemented secure authentication using JWT tokens and AES-256 encryption for patient data, ensuring compliance with HIPAA data handling standards

MeetNow: Real-time Video Conferencing Platform, *Next.js, Node.js, Mediasoup, WebRTC, PostgreSQL, Docker*

- Constructed a scalable video conferencing platform (Google Meet clone) using Next.js frontend and Node.js backend, implementing Mediasoup SFU (Selective Forwarding Unit) architecture to support 50 concurrent participants per room
- Orchestrated low-latency audio and video streaming using WebRTC protocols with STUN/TURN servers, maintaining stable peer-to-peer connections with average latency below 150 milliseconds under concurrent multi-user workloads
- Automated cloud deployment workflows using Docker containerization and CI/CD pipelines with GitHub Actions, reducing application release cycles by 40 percent

IQuantum: Educational Platform, *React.js, Node.js, Express.js, Mongodb*

- Developed a comprehensive educational platform featuring separate Frontend, Backend, and Admin Panel modules.
- Built optimized RESTful APIs using Node.js and Express.js with PostgreSQL database, reducing SQL query execution times by 30 percent through query optimization and database indexing.

GROVE: Geospatial Travel Discovery Platform, *Java, Spring Boot, PostgreSQL*

- Engineered a location-based travel discovery platform using Java and PostgreSQL.
- Optimized spatial query performance by implementing R-tree indexing and query execution plan optimization, reducing multi-parameter location search latency by 60 percent.

Certifications and Achievements

AWS Academy Graduate, *Cloud Foundations and Data Engineering*

Infosys, *Artificial Intelligence Primer*

NPTEL, *Certified in Software Project Management*

Smart India Hackathon 2024,
Led a 5-member backend development team to a top 10 finish among 500 participating teams nationwide

Pearson English International Certificate,
Level 10 (C1 Advanced Proficiency in professional English communication)

Languages

English: Professional Working Proficiency

Telugu: Native Speaker

Hindi: Conversational Proficiency